

Lab Written Exam

Topics

- Network Cabling
 - HTTP / DNS / FTP Server Configuration
 - VLAN Configuration
 - Static Routing Configuration
 - RIP Routing Configuration
 - Wireless Routing Configuration (One Hop)
-

Network Cabling

a) Difference between Straight-through and Crossover Cable

Feature	Straight-through Cable	Crossover Cable
Wiring	Same color order on both ends	Different color order on ends
Purpose	Connect different devices	Connect similar devices
Example	PC → Switch	Switch → Switch, PC → PC
Signal	No swapping	Transmit & Receive are swapped

b) T568A and T568B Standards

These are wiring standards used for arranging wires inside Ethernet cables.
They define the color code sequence of 8 wires.

T568A Color Order	T568B Color Order
White-Green	White-Orange
Green	Orange
White-Orange	White-Green
Blue	Blue
White-Blue	White-Blue
Orange	Green
White-Brown	White-Brown
Brown	Brown

Difference: Only orange and green pairs are swapped.

c) Cable Usage

- PC → Switch → Straight-through cable
- Switch → Switch → Crossover cable

d) Purpose of Network Cabling

The purpose of **network cabling** is to connect different network devices and enable communication between them.

It allows devices such as computers, switches, routers, and servers to **transmit data and share resources efficiently**.

Main purposes:

- **Data transmission** → Sends and receives data
- **Device connectivity** → Connects network devices
- **Resource sharing** → Files, printers, internet
- **Reliable communication** → Stable and secure connection
- **Network performance** → Fast and efficient transfer

Conclusion:

Network cabling forms the **physical backbone of a network**.

HTTP / DNS / FTP Server Configuration

a) What is HTTP Server?

An **HTTP (HyperText Transfer Protocol) server** stores and delivers web pages to clients (browsers).

- Uses **HTTP protocol**
- Default port: **80**
- Sends requested web pages to browsers

Examples: Apache, Nginx

Used for **hosting websites and web applications**.

b) What is DNS and how it works?

DNS (Domain Name System) converts domain names into IP addresses.

Example:

www.google.com → 142.x.x.x

Working Process:

1. User enters domain name
2. DNS receives request
3. Searches for IP address
4. Returns IP to browser
5. Browser connects to server

DNS eliminates the need to remember IP addresses.

c) What is FTP and its uses?

FTP (File Transfer Protocol) transfers files between computers.

- Default port: **21**

Uses:

- Upload files
- Download files
- Share data
- Manage website files

Commands:

- `put` → upload
- `get` → download

VLAN Configuration

a) What is VLAN and why is it used?

A **VLAN (Virtual Local Area Network)** logically groups devices regardless of physical location. It divides a network into multiple **broadcast domains**.

Uses:

- Improves security
- Reduces broadcast traffic
- Better network management
- Enhances performance

b) VLAN 10 Configuration

```
enable
configure terminal
vlan 10
name VLAN10
exit
interface fastethernet 0/1
switchport mode access
switchport access vlan 10
exit
```

c) What is the difference between access port and trunk port?

Feature	Access Port	Trunk Port
VLAN support	Single VLAN	Multiple VLANs
Traffic type	Untagged	Tagged
Usage	Connect end devices (PCs)	Connect switches/routers
Configuration	Simple	More complex

Conclusion

Access ports are used for end devices, while trunk ports are used to carry traffic between network devices across multiple VLANs.

Static Routing Configuration

1. What is static routing, why we use it, and how it works

Answer:

Static routing is a method of routing where routes are **manually configured** by the network administrator in the routing table.

Why it is used:

- Suitable for small networks
- Provides better security (no automatic updates)
- Gives full control over routing
- Uses less bandwidth (no routing updates)

How it works:

- The administrator enters a route manually using commands
- The router stores this route in its routing table
- When a packet arrives, the router checks the destination network
- It forwards the packet to the specified **next-hop address**

RIP Routing Configuration

2. What is RIP and how it works

Answer:

RIP (Routing Information Protocol) is a dynamic routing protocol that automatically updates routing tables.

- It uses **hop count** as a metric
- Maximum hop count is **15**

How it works:

- Routers share their routing tables with neighbors every 30 seconds

- Each router calculates the best path based on **minimum hop count**
 - If a route changes, it is automatically updated
 - If hop count exceeds 15, the network cannot be reached
-

3. Difference between Static Routing and Dynamic Routing

Feature	Static Routing	Dynamic Routing
Configuration	Manual	Automatic
Updates	No updates	Updates automatically
Complexity	Simple	Complex
Scalability	Not suitable for large networks	Suitable for large networks
Bandwidth usage	Low	Higher (due to updates)

4. When would you use static routing instead of RIP?

Answer:

Static routing is used instead of RIP in the following cases:

- In **small networks** with few routers
- When **network paths do not change frequently**
- When **security is important** (no route advertisements)
- When **low bandwidth usage** is required
- When **simple configuration** is preferred

Wireless Routing Configuration

1. What is wireless routing and how it works

Answer:

Wireless routing means connecting devices to a network using **Wi-Fi instead of cables**.

A **wireless router** is used to send and receive data.

How it works:

- The router is connected to the internet
 - It sends signals through the air (Wi-Fi)
 - Devices connect using **SSID (Wi-Fi name) and password**
 - Data is sent from device → router → internet
-

2. Advantages of wireless routing

Answer:

- No need for cables
- Easy to install
- Users can move freely
- Can connect many devices
- Saves time and cost

Why is **192.168.1.0** used in static routing (why 0 at the end)?

Answer:

192.168.1.0 is a **network address**, not a host address.

- In a subnet like **192.168.1.0/24**, the last part (**.0**) represents the **network ID**
- It identifies the whole network, not a specific device
- Routers use the **network address** to route traffic to that entire network

Example:

- Network: **192.168.1.0**
- Usable IPs: **192.168.1.1 – 192.168.1.254**
- Broadcast: **192.168.1.255**

So, we use **.0** because it represents the **entire network**.

Which is better: Static Routing or RIP? Why?

Answer:

It depends on the network size and requirement.

Static Routing is better when:

- The network is **small**
- Routes do not change often
- **Security** is important
- You want **full control**
- You want to save **bandwidth**

RIP is better when:

- The network is **large**
- Routes change frequently
- You want **automatic updates**
- Easy to manage without manual configuration

Conclusion:

- **Static routing** is better for **small and secure networks**
- **RIP** is better for **large and dynamic networks**